

Clustered Safety Events and the Effective Sample Size in Autonomous Vehicle Fleet Rate Estimation

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Abstract

Autonomous vehicle fleet validation estimates rare-event rates — safety-relevant disengagements, near-misses, contact events — per mile driven, and attaches confidence statements to those estimates. The estimators in current use treat logged events as independent arrivals of a Poisson process. Real fleet events are not independent: a construction zone, a rain hour, or a single poorly handled intersection produces several correlated events. We model the event stream as a compound Poisson (Neyman–Scott) process and show that ignoring this clustering inflates the variance of the rate estimator by a design effect $D = E[K^2]/E[K]$, where K is the cluster size. Three consequences follow. First, a nominal 95% confidence interval has asymptotic coverage $2\Phi(z_{0.975}/\sqrt{D}) - 1$, which falls to 74% at $D = 3$ and 54% at $D = 7$. Second, the mileage required to demonstrate that a fleet’s event rate lies below a safety target is understated by exactly the factor D , which suggests that standard mileage calculations may understate the evidence required. Third, empirical-Bayes shrinkage weights of the form $n/(n + k)$ over-weight noisy local estimates, costing up to $7.8\times$ in mean squared error in our study. We give a correction that substitutes an effective sample size $n_{\text{eff}} = n/\hat{D}$, where $\hat{D}$ is estimated from the logs themselves, and show by simulation that it restores nominal coverage across the range studied. We then measure $\hat{D}$ on public California DMV autonomous vehicle disengagement reports. Pooled across fleets it is 4.79 for 2023 and 5.29 for 2024, and per-fleet estimates range from 1.00 to 13.77 — so the design effect is neither negligible nor a constant that can be assumed. As a negative control we repeat the measurement on NHTSA Standing General Order driverless incident reports, where the vehicle-day design effect is 1.02: the estimator returns unity when events genuinely do not recur on the clustering unit, which both supports the larger values and scopes the correction to the event classes that do recur. **The simulation study uses synthetic data. The measured design effects come from regulatory filings, and the two sources describe different event classes — safety-driver disengagements and driverless reportable incidents — which is the distinction the negative control turns on.** This is an independent preprint and has not been peer reviewed.

1 Introduction

The central statistical difficulty of autonomous vehicle (AV) safety validation is that the events of interest are rare. Kalra and Paddock [5] showed that demonstrating a fatality rate lower than the human benchmark with reasonable confidence would require hundreds of millions of miles, a result that has shaped a decade of subsequent work. Liu and Feng [8] give the modern statement of the problem as a “curse of rarity”: the information content per mile is so low that direct estimation is impractical without variance reduction, and much recent effort has gone into importance sampling and scenario-based testing to recover statistical power.

Much of this work rests on a modelling assumption that is seldom made explicit and, as far as we are aware, not yet examined directly: that the safety-relevant events logged by a fleet are

independent. Terres et al. [13] model the count of behavioral events over m miles as $\text{Poisson}(\theta m)$, which embeds independence in the likelihood. Chen et al. [2] assume i.i.d. features and a Poisson sample size in deriving confidence intervals for importance-sampled rate estimates; the authors note explicitly that “how to properly design the run segments to minimize spatial-temporal dependencies is beyond the scope of the current paper.” Zhao et al. [14], in a paper arguing for exactly the statistical rigour we are concerned with here, adopt i.i.d. Bernoulli scenario models while acknowledging that the simplification “may not hold due to factors like software updating of the AV or changing environments.”

The independence assumption is analytically convenient and in many settings harmless. In this one, we argue, it is unlikely to hold, in a specific and consequential way. Safety-relevant events arrive in clusters. A construction zone that the perception stack handles poorly produces several disengagements in a few minutes. A rain hour degrades detection across an entire shift. One unprotected left turn geometry generates the same near-miss on every pass. The clusters are not a nuisance detail of the logging pipeline; they are how the underlying failure modes express themselves.

This paper makes the consequence precise. We model the event stream as a compound Poisson process, derive the resulting variance inflation, and quantify what it costs. The contribution is deliberately narrow:

1. We show that the naive Poisson rate estimator remains unbiased but that its variance is inflated by a design effect $D = E[K^2]/E[K]$ (Section 3).
2. We give the exact asymptotic coverage of the naive confidence interval as a function of D , and confirm it by simulation (Proposition 1).
3. We show that the mileage required to demonstrate a safety target scales linearly in D (Proposition 2), which is the safety-critical consequence: a fleet using the standard calculation concludes “demonstrated” before the evidence supports it.
4. We give a correction based on an effective sample size that is estimable from fields fleets already log, and show it restores nominal coverage (Sections 4 and 5).
5. We measure $\hat{D}$ on public California DMV disengagement filings for 2023 and 2024, covering twelve fleets, and find pooled values near 5 with per-fleet values spanning 1.00 to 13.77 (Section 6).
6. We repeat the measurement on NHTSA driverless incident reports as a negative control, obtain $\hat{D} = 1.02$ at the vehicle-day, and use it to scope the claim (Section 6.3).

We are explicit about what this is not. The design effect is textbook survey statistics [6, 3], and cluster-robust variance estimation is standard [7]. No new mathematics is claimed. The contribution is to connect that machinery to a setting where it has not yet been applied, to quantify what its absence costs in the regime AV validation operates in, and to note that the correction requires no new instrumentation.

2 Related Work

Fleet-level rate estimation. The line of work most directly affected is the estimation of event rates from fleet logs. Terres et al. [13] develop maximum likelihood procedures for rate estimation under partial event review, using a Poisson model for the event count. Chen et al. [2] construct

confidence intervals for importance-sampled rate estimates, assuming independent features. Zhao et al. [14] propose a probability-of-failure-per-scenario metric and i.i.d. Bernoulli models for scenario-based testing. None of these addresses clustering, serial dependence, or effective sample size directly. Two of the three are explicit that dependence lies outside their scope, and it is that explicitness which makes the present extension straightforward to state.

Traffic conflict extremes. A separate and considerably more dependence-aware literature estimates crash risk from traffic conflicts using extreme value theory, beginning with Songchitruksa and Tarko [12]. Zheng and Sayed [15] develop non-stationary Bayesian hierarchical peak-over-threshold models, and more recent work in the same literature extends conditional peak-over-threshold models with self- and cross-exciting structure and with spatio-temporal hierarchies. This machinery explicitly models dependence between extreme observations.

It does not, however, transfer directly to the fleet setting, and the reason matters for the present paper. Self-exciting and conditional peak-over-threshold models describe a single, densely observed excitation stream at one site: an intersection under continuous video observation, where every conflict is recorded and the excitation kernel is identifiable from inter-event times. Fleet validation aggregates sparse events across many routes, operating conditions, and software releases, where the dependence is *nested* — an event sits within a scene, within a drive, within a release — rather than generated by one excitation process in one time series. Nested dependence of this kind is what design effects and cluster-robust variance estimators are built for, and a point-process excitation model is not the natural description of it.

Dependence corrections elsewhere. Extreme value theory has long handled dependent exceedances by declustering, with the extremal index quantifying the degree of clustering [4]. In survey statistics the design effect [6, 3] plays the same role for clustered sampling designs, and generalized estimating equations [7] provide cluster-robust variance estimates for correlated observations. In financial machine learning, López de Prado [9] identifies overlapping event labels as a source of inflated effective sample size and proposes average-uniqueness weighting and sequential bootstrapping as remedies. The present paper connects these adjacent treatments to the fleet rate estimation setting, where the same difficulty arises but the corresponding corrections have not yet been carried over.

3 Problem Formulation

3.1 A clustered event process

Let a fleet accumulate m miles. We model safety-relevant events as a compound Poisson process of Neyman–Scott type [11]. Let cluster centres — distinct difficult contexts, such as one construction zone encountered once, one rain hour, one problematic intersection geometry — arrive as a Poisson process with rate λ_c per mile, so that the number of clusters is $N_c \sim \text{Poisson}(\lambda_c m)$. Each cluster i contributes K_i logged events, with $K_1, K_2, \dots$ independent and identically distributed on $\{1, 2, \dots\}$, independent of N_c . The total event count is

$$N = \sum_{i=1}^{N_c} K_i. \quad (1)$$

The quantity validation cares about is the event rate per mile,

$$\theta = \lambda_c E[K], \quad (2)$$

estimated by $\hat{\theta} = N/m$. The classical Poisson model is the special case $K \equiv 1$.

3.2 The design effect

By the standard compound Poisson moment identities,

$$E[N] = \lambda_c m E[K], \quad \text{Var}(N) = \lambda_c m E[K^2]. \quad (3)$$

The estimator $\hat{\theta} = N/m$ is therefore unbiased for θ , with

$$\text{Var}(\hat{\theta}) = \frac{\lambda_c E[K^2]}{m} = \frac{\theta}{m} \cdot \frac{E[K^2]}{E[K]} = \frac{\theta D}{m}, \quad D := \frac{E[K^2]}{E[K]}. \quad (4)$$

We call D the design effect, by analogy with clustered survey sampling [6]. It satisfies $D \geq 1$, with equality if and only if every cluster has size one. Writing $\mu_K = E[K]$, an equivalent form is $D = \mu_K + \text{Var}(K)/\mu_K$: clustering hurts both through the mean cluster size and through its dispersion.

An analyst who assumes independence uses $\text{Var}(\hat{\theta}) = \theta/m$ and therefore understates the standard error by a factor of $\sqrt{D}$. Note that the point estimate is unaffected. The damage is entirely to the uncertainty statement — which, in a safety argument, is the part that carries the claim.

Remark 1. Equation (4) is the perfectly-correlated case of Kish's design effect $D = 1 + (\bar{K} - 1)\rho$, corresponding to $\rho = 1$: every event in a cluster is treated as carrying the information of the same underlying failure mode. A process with partial within-cluster correlation interpolates between $D = 1$ and the value in (4), so the figures reported here should be read as the design effect implied by whatever clustering the analyst's own logs exhibit, not as a universal constant.

3.3 Coverage of the naive interval

Proposition 1 (Undercoverage). *Let $\hat{\theta} = N/m$ and let the naive interval be $\hat{\theta} \pm z_{1-\alpha/2} \sqrt{\hat{\theta}/m}$. As $m \rightarrow \infty$ with θ and D fixed, the asymptotic coverage of this interval is*

$$2\Phi\left(\frac{z_{1-\alpha/2}}{\sqrt{D}}\right) - 1, \quad (5)$$

where Φ is the standard normal distribution function.

Proof. By the central limit theorem for compound Poisson sums, $(\hat{\theta} - \theta)\sqrt{m/(\theta D)} \Rightarrow \mathcal{N}(0, 1)$. The naive half-width is $z_{1-\alpha/2} \sqrt{\hat{\theta}/m}$, and $\hat{\theta} \rightarrow \theta$ in probability, so the half-width is asymptotically $z_{1-\alpha/2} \sqrt{\theta/m}$, which equals $z_{1-\alpha/2}/\sqrt{D}$ true standard errors. Coverage is therefore the probability that a standard normal variate lies within $z_{1-\alpha/2}/\sqrt{D}$ of zero. $\square$

For a nominal 95% interval this gives 87.1% at $D = 1.67$, 74.2% at $D = 3$, and 54.1% at $D = 7$. At the upper end of that range the interval's actual coverage bears little relation to its nominal level.

3.4 Miles required to demonstrate a target

Validation programmes are usually framed as a demonstration: show, with confidence $1-\alpha$, that the fleet’s event rate lies below some target $\theta^* > \theta$. Using the upper confidence limit $\hat{\theta} + z_{1-\alpha/2} \sqrt{\text{Var}(\hat{\theta})}$ and solving for the mileage at which it falls below θ^* gives the following.

Proposition 2 (Required mileage). *Under the clustered process, the mileage required for the upper confidence limit to fall below θ^* is*

$$m_{\text{req}} = \frac{z_{1-\alpha/2}^2 \theta D}{(\theta^* - \theta)^2}, \quad (6)$$

which exceeds the mileage computed under the independence assumption by exactly a factor of D .

This is the statement with the most direct practical consequence. For design effects in the range studied here, a programme that sizes its mileage under the independence assumption and stops at that figure will have accumulated between one third and one seventh of the evidence implied by its own confidence statement. We note that the discrepancy runs in the less cautious direction.

4 Proposed Correction

The correction is to estimate D from the logs and use it. Suppose the N logged events can be partitioned into C observed clusters with sizes $K_1, \dots, K_C$. A natural moment estimator of the design effect is

$$\hat{D} = \frac{\sum_{i=1}^C K_i^2}{\sum_{i=1}^C K_i}, \quad (7)$$

which is the sample analogue of $E[K^2]/E[K]$. The effective sample size is then

$$n_{\text{eff}} = \frac{N}{\hat{D}}, \quad (8)$$

and the corrected interval is

$$\hat{\theta} \pm z_{1-\alpha/2} \frac{\sqrt{N\hat{D}}}{m} = \hat{\theta} \pm z_{1-\alpha/2} \frac{\sqrt{N}}{m} \sqrt{\hat{D}}. \quad (9)$$

The required-mileage formula of Proposition 2 is corrected by the same factor.

Shrinkage weights. The same substitution repairs a second estimator. Sparse per-segment or per-scenario rates are commonly stabilised by shrinking them toward a fleet-level prior with an empirical-Bayes weight $w = n/(n+k)$, so that a segment with few events leans on the prior and a segment with many events leans on its own data. Under clustering, n overstates the independent information available, w is therefore too large, and the estimator leans on a noisier local estimate than the evidence supports. Substituting $w_{\text{eff}} = n_{\text{eff}}/(n_{\text{eff}} + k)$ restores the intended behaviour. This is the same defect López de Prado [9] identifies for overlapping labels in financial machine learning, arriving here by a different route.

What counts as a cluster. The estimator (7) requires that cluster membership be observable. In fleet logging it usually is, at least approximately, because every logged event already carries identifiers that induce a natural grouping: the vehicle and the calendar date, the drive or run segment, the scene or map location, the weather condition, or the software release. Any of these defines a partition; the appropriate choice is the one matching the failure mode under study. We return to this in Section 6.

5 Simulation Study

Setup. We simulate the process of Section 3 with a true rate of $\theta = 10^{-4}$ events per mile (one per ten thousand miles) over a 5,000,000 mile deployment, with 20,000 replications per configuration. Cluster sizes are drawn from a geometric distribution on $\{1, 2, \dots\}$ with parameter p , giving mean cluster size $1/p$ and design effect $D = (2 - p)/p$. All data are synthetic; the accompanying script `sim_clustered_av_rates.py` reproduces every figure reported here.

Coverage. Table 1 reports the actual coverage of a nominal 95% interval. The simulated coverage of the naive interval agrees with Proposition 1 to within one percentage point throughout, and the corrected interval holds its nominal level across the whole range. Figure 1 shows the same result over a finer grid.

Table 1: Actual coverage of a nominal 95% confidence interval for the fleet event rate. Synthetic data; 20,000 replications per row.

Mean cluster size	D	Naive (sim.)	Naive (theory)	Corrected	CI width ratio
1.00	1.00	94.8%	95.0%	94.8%	1.00×
1.33	1.67	86.9%	87.1%	94.9%	1.29×
2.00	3.00	73.2%	74.2%	94.5%	1.73×
2.86	4.71	63.0%	63.3%	94.4%	2.17×
4.00	7.00	53.6%	54.1%	94.6%	2.64×

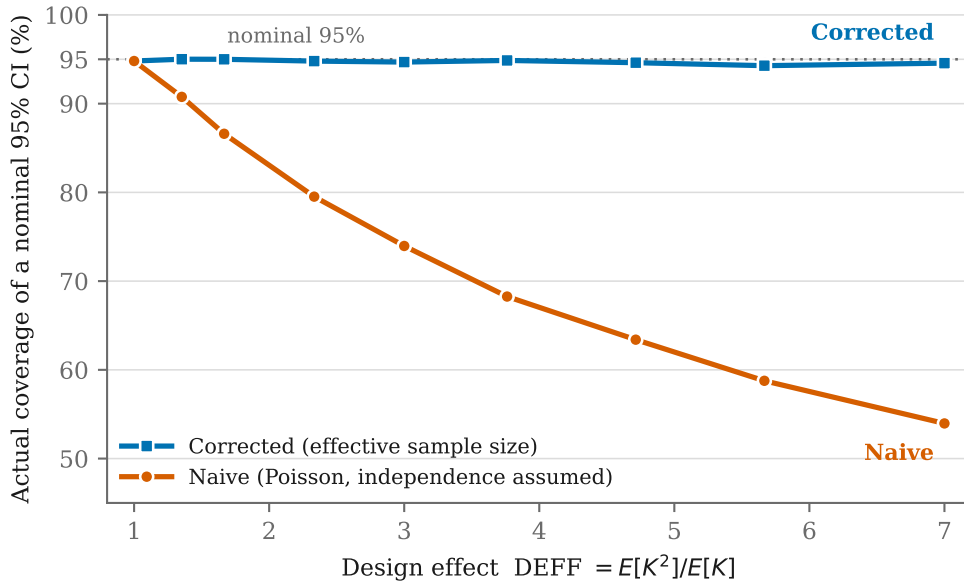


Figure 1: Actual coverage of a nominal 95% confidence interval as a function of the design effect. The naive interval degrades steadily; the effective-sample-size correction holds the nominal level. Synthetic data.

A mean cluster size of 2 — two logged events per difficult context, which is not an extreme value for fleet data — already reduces a nominal 95% interval to 73% actual coverage.

Required mileage. Table 2 reports the mileage needed for the upper confidence limit to fall below a target of 1.5θ . The ratio is exactly D , as Proposition 2 requires.

Table 2: Mileage required to demonstrate that the event rate lies below $\theta^* = 1.5\theta$ at 95% confidence, with $\theta = 10^{-4}$ per mile.

Mean cluster size	D	Assuming independence	Corrected	Shortfall
1.00	1.00	153,658	153,658	$1.00\times$
1.33	1.67	153,658	256,097	$1.67\times$
2.00	3.00	153,658	460,975	$3.00\times$
2.86	4.71	153,658	724,389	$4.71\times$
4.00	7.00	153,658	1,075,608	$7.00\times$

Shrinkage. Table 3 reports the mean squared error of a per-segment rate blended toward an unbiased fleet-level prior with $k = 15$, over 200,000 miles. Because the prior is unbiased by construction, the entire gap is variance: using the nominal event count costs up to $7.8\times$ in mean squared error relative to using the effective count.

Table 3: Mean squared error of an empirical-Bayes blended per-segment rate, using the nominal event count n versus the effective count $n_{\text{eff}} = n/\hat{D}$, with $k = 15$.

Mean cluster size	D	MSE, nominal n	MSE, effective n	Ratio
1.00	1.00	1.647×10^{-10}	1.647×10^{-10}	$1.00\times$
1.33	1.67	2.766×10^{-10}	1.757×10^{-10}	$1.57\times$
2.00	3.00	5.172×10^{-10}	1.785×10^{-10}	$2.90\times$
2.86	4.71	8.658×10^{-10}	1.774×10^{-10}	$4.88\times$
4.00	7.00	1.292×10^{-9}	1.655×10^{-10}	$7.81\times$

6 Design Effects Measured on Public Fleet Data

Nothing in Section 4 requires new instrumentation, and this is the practical argument for the correction: $\hat{D}$ is computable from records that already exist, including public ones.

California DMV disengagement reports. Manufacturers testing autonomous vehicles on California public roads file annual disengagement reports [1]. The published CSV files contain one row per disengagement, with fields for the manufacturer, the date, the vehicle identification number, whether a driver was present, what initiated the disengagement, the road type, and a free-text description of the cause. The date and vehicle identifier together define a vehicle-day, which is a natural and conservative cluster unit: events logged by the same vehicle on the same day are candidates for sharing an underlying context.

6.1 Measured design effects

We computed (7) over vehicle-day clusters for the 2023 and 2024 testing-with-a-driver filings, retaining fleets with at least 30 reported events in a year. This covers 6,562 events across 2,989

vehicle-days in 2023 and 1,773 events across 981 vehicle-days in 2024. Rows with blank manufacturer, date, or vehicle identifier were excluded; in both files every such row was entirely empty, an artefact of the export, so no reported event was discarded. Results are in Table 4.

Table 4: Design effects estimated from California DMV disengagement reports, using the vehicle-day as the cluster unit. Fleets with fewer than 30 reported events in a year are shown as “–”. These are safety-driver disengagements, not driverless safety events; see the caveats below.

Fleet	2023		2024	
	Events	$\hat{D}$	Events	$\hat{D}$
Apple	3,194	3.44	285	1.60
Ghost Autonomy	1,034	7.73	351	13.77
aiMotive	708	8.02	188	8.31
Motional AD	593	5.25	320	2.84
Bosch	314	6.37	77	5.88
Waymo	212	1.04	244	1.19
Qualcomm Technologies	197	1.65	–	–
IMAGRY	124	2.81	136	4.44
Nuro	47	1.13	103	1.10
Aurora Operations	47	12.53	–	–
Woven by Toyota	42	1.00	–	–
Zoox	–	–	34	1.00
Pooled, all fleets	6,562	4.79	1,773	5.29

Three observations. First, the pooled design effect is close to 5 in both years, near the upper end of the range explored in Section 5; at that value a nominal 95% interval covers roughly 62%, and required mileage is understated fivefold. Second, and more important, $\hat{D}$ is *not* a constant: it spans 1.00 to 13.77 across fleets. Some fleets — Waymo, Nuro, Woven by Toyota, Zoox — report events that are close to unclustered, and for them the independence assumption is nearly harmless. Others — Aurora, Ghost Autonomy, aiMotive — show design effects above 8, where the naive interval is badly optimistic. This is the strongest argument in the paper for measuring D rather than assuming any value for it, including the values used in our own simulation. Third, several fleets are stable across years (aiMotive 8.02 $\rightarrow$ 8.31, Bosch 6.37 $\rightarrow$ 5.88, Waymo 1.04 $\rightarrow$ 1.19, Nuro 1.13 $\rightarrow$ 1.10), which suggests $\hat{D}$ is picking up something persistent about how a given programme operates and logs rather than year-specific noise; others move substantially (Apple 3.44 $\rightarrow$ 1.60, Ghost 7.73 $\rightarrow$ 13.77), and we cannot say from these data whether that reflects changed operations or changed reporting.

6.2 Caveats on these figures

These estimates should not be read as design effects for driverless deployments, for four reasons.

1. **Event class.** These are disengagements during testing with a safety driver. A safety driver’s interventions are not the same object as a driverless system’s safety-relevant events, and we have no basis here for assuming the clustering transfers.
2. **Reporting practice.** Filing conventions differ across manufacturers and years. A fleet reporting at coarser granularity would show a lower $\hat{D}$ for reasons having nothing to do with

its driving. We cannot distinguish this from genuinely unclustered events, which is a specific concern for the fleets reporting $\hat{D} \approx 1$.

3. **Cluster unit.** The vehicle-day is one defensible choice. Grouping instead by road type, by location, or by cause text would give different partitions and different design effects. The appropriate unit depends on the failure mode under study.
4. **Coverage.** Twelve fleets in two years is an illustrative sample, not a census, and the fleets with the largest event counts dominate the pooled figure.

What the table does establish is narrower and, we think, sufficient for the paper’s purpose: design effects substantially greater than one occur in real regulatory filings of autonomous vehicle events, they vary widely across programmes, and they are computable from data that already exists.

6.3 A negative control: driverless incident reports

The California filings describe disengagements under a safety driver. Incident reports filed under NHTSA Standing General Order 2021-01 [10] cover driverless operation, and so speak more directly to the event class of interest. We applied the same procedure to them.

Two deduplication steps are necessary first, because the SGO file is a report log rather than an event log. An incident is amended over time, producing several rows that share a Report ID at increasing Report Version; and one physical event may be reported by more than one entity, which NHTSA links with a Same Incident ID. Counting raw rows would measure how often entities file, not how often events occur. Of 1,365 rows, 1,324 remain after keeping the latest version of each report and 1,323 after collapsing shared incident identifiers.

At the vehicle-day the design effect is essentially one (Table 5).

Table 5: Design effects from NHTSA SGO driverless incident reports, vehicle-day clusters, after deduplication.

Reporting entity	Events	Clusters	Mean K	$\hat{D}$
Waymo	1,117	1,107	1.01	1.02
Avride	72	70	1.03	1.06
Zoox	61	61	1.00	1.00
Pooled	1,323	1,311	1.01	1.02

Driverless reportable incidents do not recur on the same vehicle within a day. We regard this as a negative control rather than a disappointment. It shows the estimator returns unity when events genuinely do not recur on the unit of clustering, which is evidence that the values between 5 and 13 in Table 4 reflect structure in the data rather than an artefact of the procedure. It also scopes the paper’s claim: the correction matters for event classes that recur on a unit — disengagements, near-misses, and the behavioural events that the fleet rate estimation literature actually models [13] — and not for the rarest driverless crash events at the vehicle-day level.

An apparent effect that is not one. Aggregating instead by reporting entity, city and day gives a ratio of 13.03 pooled, and 14.82 for Waymo alone. We do not report these as design effects, and the reason is instructive. A city-day contains every vehicle the operator ran in that city that

day, so a cluster of five incidents may be five independent vehicles rather than five correlated events. The data settle the question directly: Waymo’s city-days average 4.75 incidents across 4.71 distinct vehicles, and Avride’s 4.00 across 3.89. Almost every incident in a city-day comes from a different vehicle, so the apparent clustering is fleet size — exposure — and not shared context. Separating genuine city-level dependence from exposure would require vehicles or miles operated per city-day, which the SGO filings do not report. We record the ratio only as an upper bound, and note it as a trap for anyone applying this method to aggregated data without an exposure denominator.

7 Discussion and Limitations

What this establishes. That the independence assumption commonly used in AV fleet rate estimation is not without cost; that this cost is a computable function of the clustering in the data; and that the correction is available at no instrumentation cost. The point estimate is unaffected throughout — what changes is the confidence attached to it, and the mileage at which a demonstration can honestly be declared complete.

Scope of the claim. The two measurements together bound what is being claimed more tightly than either would alone. Design effects well above one occur in real filings, vary widely between programmes, and are computable from existing data — but they are a property of an event class and a clustering unit, not of autonomous vehicles in general. For driverless reportable incidents at the vehicle-day, $\hat{D} = 1.02$ and the independence assumption is adequate. The correction earns its keep for the event classes that recur on a unit, which is the class the fleet rate estimation literature actually models [13, 2]. What D equals for any particular programme and event definition remains a question that programme has to answer from its own logs; the contribution here is that the question is answerable and currently unasked.

What this does not establish. Every number in Section 5 is generated from an assumed cluster-size distribution rather than a fitted one, so the simulation demonstrates the mechanism rather than measuring the world. And the two data sources are observational: we can measure how reported events cluster, but not why, and in particular we cannot separate genuine operational clustering from clustering induced by how a given programme chooses to file.

Direction of the error. Under the process modelled here, ignoring clustering makes uncertainty statements overconfident, which is the unsafe direction. We emphasise that this follows from the model, not from a general principle: a process in which logging systematically *suppresses* repeated events within a context, for example through deduplication in the event detection pipeline, could push in the opposite direction. Analysts should determine the sign for their own pipeline rather than assume it.

Cluster identifiability. The correction assumes cluster membership is observable. Where scene or segment boundaries are ambiguous — and they often are, since “the same difficult context” is not a field any logger writes directly — $\hat{D}$ is itself estimated with error, and we do not propagate that uncertainty into the interval. A fully Bayesian treatment placing a prior over the partition would address this and is left to future work. A pragmatic interim measure is sensitivity analysis across several plausible cluster definitions.

Model scope. We assume cluster sizes are independent and identically distributed and independent of the cluster arrival process. Real fleets violate this: difficult contexts are more common in some operational design domains than others, and a software release changes both the arrival rate and the size distribution at once. A hierarchical model with release-level and domain-level effects is the natural extension.

Relationship to existing practice. Fleets are of course aware that their data is structured; engineering teams routinely analyse events by scene, by route, and by release. Our claim is narrower: that this structure does not appear to be propagated into the published confidence statements and mileage targets, and that propagating it changes those statements materially. We would be glad to be shown otherwise for any particular programme.

Acknowledgments

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